Srinivasa Sai Chava

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Research Interests

Multilingual and low-resource natural language processing; code-mixed content moderation; retrieval-augmented generation and LLM evaluation; multimodal learning and robustness.

Education

Indian Institute of Information Technology Dharwad May 2025
B.Tech. in Data Science and Artificial Intelligence CPI: 7.93/10.00
Selected grades: Machine Learning A; Natural Language Modeling: ANN to LLM A−; Major Project A.

Publications and Accepted Papers

HateLens: Hierarchical Hate-Speech Detection for Tamil Code-Mixed Text
DravidianCodeMix shared task, FIRE 2026 working notes. *Accepted*

ByteBreaker@DravidianLangTech 2026: XLM-RoBERTa Large with Sliding-Window Chunking and Top-K Mean Pooling for Writing Style Classification
Chava Srinivasa Sai, R Vinay Kumar, Jigeesha Sai Surapaneni, and Chava Shanmukha Sai.
Sixth DravidianLangTech Workshop, ACL 2026, pp. 158–162. Jul 2026

Impact of Internet Derived Information Obstruction Treatment (IDIOT) Syndrome-a student cohort study
Chava Srinivasa Sai, Mandapalli Ruthvik, and Ramesh Athe.
Tuijin Jishu/Journal of Propulsion Technology, 47(01). Jan 2026

IIITDWD_SVC@DravidianLangTech-2024: Breaking Language Barriers; Hate Speech Detection in Telugu-English Code-Mixed Text
Chava Sai, Rangoori Kumar, Sunil Saumya, and Shankar Biradar.
Fourth DravidianLangTech Workshop, EACL 2024, pp. 119–123. Mar 2024

Research Experience

Machine Learning Research Assistant, IIIT Dharwad Aug 2023 – Jul 2024

- Led a team comparing Google Translate and IndicTrans2 on Hindi–English parallel text using semantic textual similarity and BLEU; analysed systematic translation errors and meaning preservation.

Industry and Teaching Experience

AI Developer and Trainer, Datavalley, Vijayawada Jan 2025 – Present
Full-time; with a break for the AGS internship, May–Aug 2026.

- Taught deep learning, NLP, and agentic AI tools through hands-on projects; mentored students in model implementation, training, evaluation, and debugging.
- Delivered Flutter and Firebase app-development instruction and guided students through application implementation.

AI / Machine Learning Intern, American Green Solutions May – Aug 2026

- Built Astra AI, a solar-operations RAG assistant with FastAPI, React, and ChromaDB, including document ingestion, source citations, streamed answers, and JWT role-based access.
- Implemented semantic FAQ caching, model fallbacks, and token-usage tracking to manage LLM cost and latency.

Selected Research Projects

HateLens: Hierarchical Tamil Hate-Speech Detection Code | Jul 2026

- Ranked **1st of 9 ranked teams** in DravidianCodeMix 2026 Subtask 1 (Tamil), with **0.4603 mean macro-F1** across five label levels on the official test set.
- Designed, trained, and evaluated a per-level ensemble of MuRIL/XLM-R encoders and QLoRA-adapted language models, with factorized classification heads and prior-based calibration for rare labels.
- Documented controlled negative results, validation-tuning limitations, and the validation-to-test generalization gap.

Faithfulness and Attribution in Retrieval-Augmented Generation Code | Mar – May 2026

- Designed and implemented a BM25/Contriever and Llama/Mistral pipeline with NLI-based claim-support scoring; compared retrieval depth, prompt styles, model choices, and inserted distractor passages.
- Manually reviewed 50 examples to examine disagreement between automated support scores and human judgments, identifying limitations of automatic evaluation.

Weakly Supervised Video Understanding Code | Apr 2026

- Built a violence-detection pipeline using frozen VideoMAE-B features and RTFM-style multiple-instance learning; achieved **0.922 mean video-level AUC** on the UCF-Crime violence subset across three seeds (s.d. 0.010).
- Evaluated a single checkpoint without retraining on XD-Violence (AUC 0.836), RWF-2000 (0.862), and ShanghaiTech (0.371), documenting the limits of cross-dataset generalization.

Honors and Awards

- **CC-MMD 2026 Grand Challenge, Task B:** joint 1st (Tamil), 1st (English), and 2nd (Chinese), team ByteBreaker.
- **DravidianCodeMix 2026, Subtask 1 (Tamil):** 1st of 9 ranked teams, team HateLens.

Technical Skills

Languages: Python, C++, Java
ML and NLP: PyTorch, scikit-learn, Hugging Face Transformers, LoRA/QLoRA, MuRIL, XLM-R, VideoMAE, experimental evaluation
LLM and Retrieval: RAG, LangChain, ChromaDB, FAISS, BM25
Application Development: FastAPI, Flask, React, Node.js, Flutter, Firebase, PostgreSQL
Tools and Platforms: Git, Docker, Google Colab Pro, AWS, Google Cloud Run

Standardized Test

GRE General: Quantitative 168/170; Verbal 156/170; Analytical Writing 3.5/6. Dec 2022